Module: Phenotyping

Identifying cell types or functional states is crucial for analyzing single-cell and spatial gene expression experiments. Researchers often use algorithms designed for non-spatial single-cell data. While these algorithms may work well for some spatial transcriptomics (ST) technologies that provide data at the single-cell level (e.g., MERFISH, CosMx, Xenium), they may not yield reliable results for multicellular-level technologies (e.g., Visium). The spatialGE app includes two methods developed for multicellular and single-cell ST data:

- **STdeconvolve (Miller et al. 2022):**
  For multi-cellular ST (e.g., Visium). This method provides estimates of cell type proportions for each spot. Proportions are estimated using Latent Dirichlet Allocation (LDA), which predicts cellular "topics". The biological identification of the topics is facilitated using gene set enrichment analysis (*see page 2*).

- **InSituType (Danaher et al. 2022):**
  For single-cell ST (e.g., CosMx), this method features a Bayes classifier, which estimates the likelihood of each cell in the ST data set of being from the same type defined in a reference expression data set (*see page 11*).

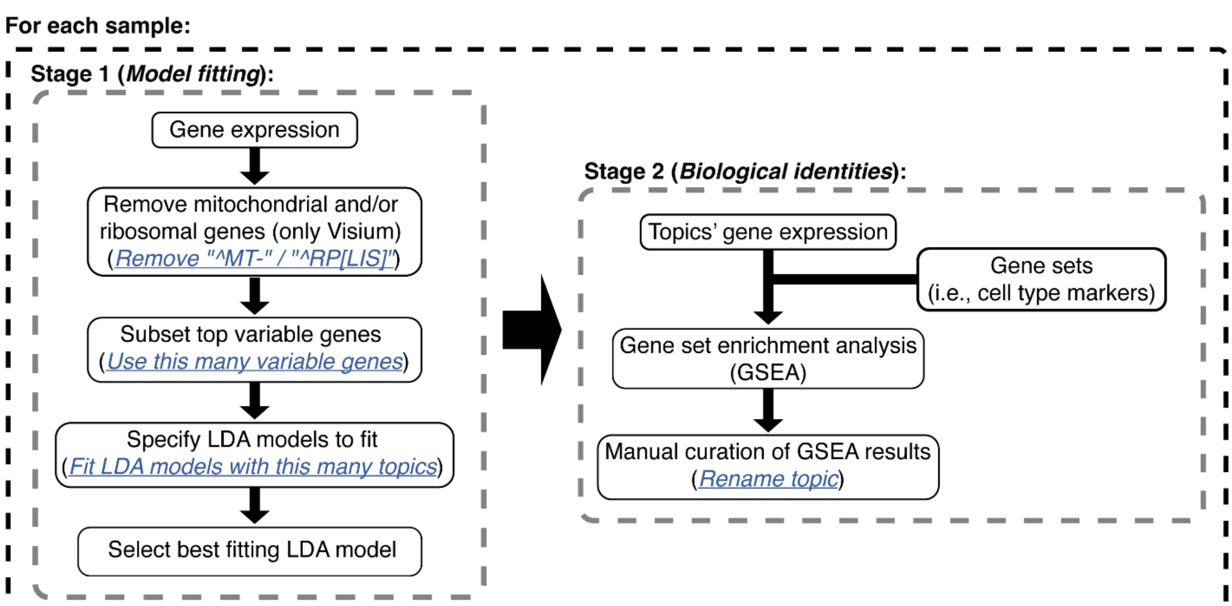
STdeconvolve

The STdeconvolve algorithm (Miller et al. 2022) has been integrated into the **Phenotyping** module in spatialGE to assign biologically meaningful classes (e.g., cell types, tissue niche types, functional states) to spots in multicellular-level spatial transcriptomics (ST) experiments.

Briefly, STdeconvolve uses Latent Dirichlet Allocation (LDA) to identify latent topics within gene expression data. These latent topics represent gene expression patterns that may correspond to cell types, tissue niches, or functional states within the tissue. To determine the biological identity of these topics, STdeconvolve includes built-in gene set enrichment analysis (GSEA). This matches combinations of highly expressed genes in the topics with genes from predefined gene sets or pathways. In the future, users will be able to upload custom gene sets.

The STdeconvolve algorithm is implemented in spatialGE as a two-stage procedure:

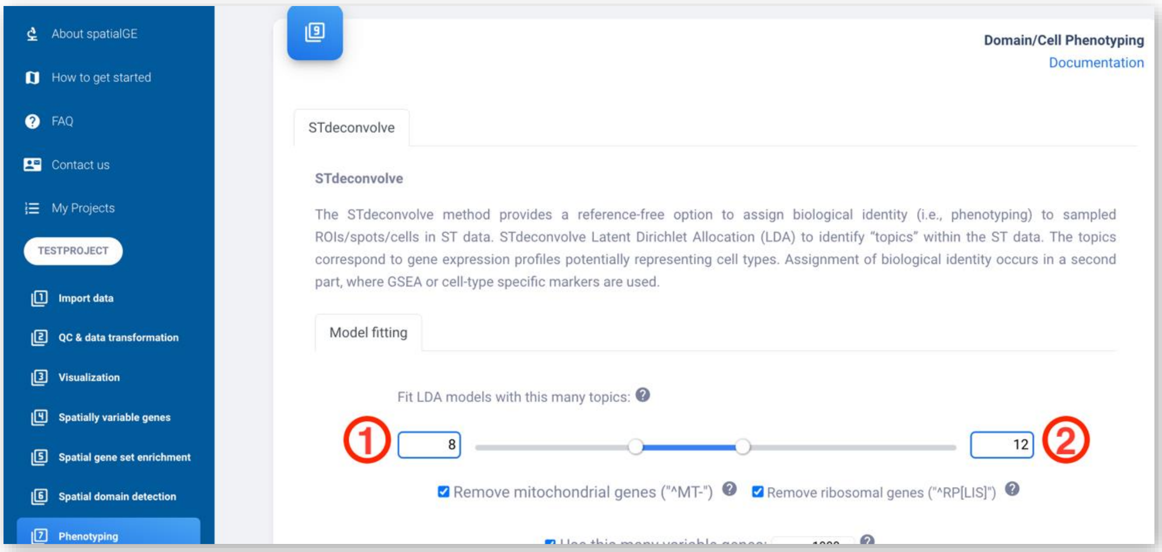
- *Stage one:* Users fit LDA models with varying numbers of topics to the gene expression data. Using specific model metrics (discussed later), the user selects the best-fitting model.
- *Stage two:* GSEA is performed to assign biological identities (e.g., cell types, tissue niche types, functional states) to each topic in the best-fitting model.



STdeconvolve – LDA Model Fitting

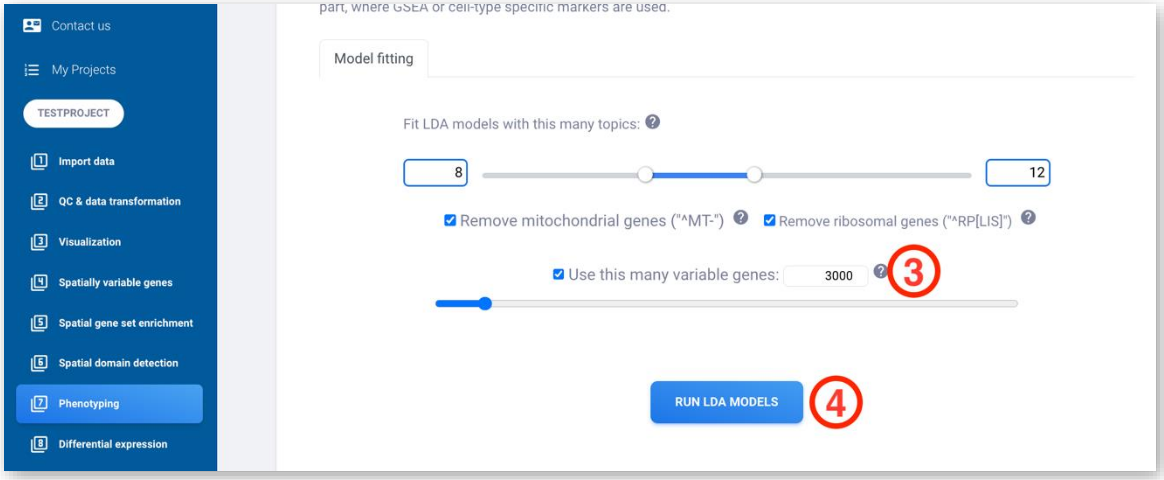
The first stage in this module is to select the number of LDA models to be fitted and the number of topics within each model. Both the number of models and topics per model are controlled by the slider labeled **Fit LDA models with this many topics**. For this example, the slider will be set with a minimum of 8 and a maximum of 12 topics. Consequently, five LDA models will be fitted: one with 8 topics, one with 9 topics, one with 10 topics, one with 11 topics, and one with 12 topics.

The number of topics defines the possible number of cell types to be detected in the sample. Testing more models will take more time to complete STdeconvolve. However, testing a wide range of models provides more options to select the LDA model that best represents the tissue biology of the sample. In general, multi-cellular ST data is limited in capturing fine-grained differences among cell subpopulations. Therefore, it is suggested to first attempt to recover the major cell types.

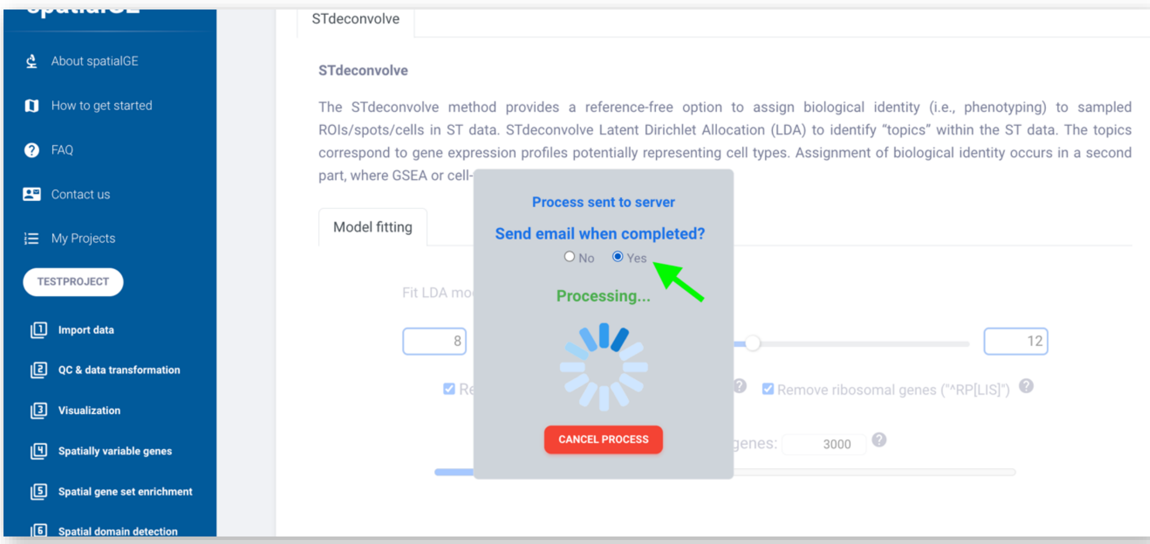


To remove mitochondrial and ribosomal genes, ensure the **Remove mitochondrial genes ("^MT-")** and **Remove ribosomal genes ("^RP[L|S]")** options are checked. The user must exercise caution, as this filter only affects data sets where mitochondrial gene names start with "MT" and ribosmomal gene names start with "RP" followed by "L" or "S" (e.g., Visium). Often, LDA models define clusters based on the high abundance of mitochondrial and/or ribosomal genes in certain tissue regions, especially diseased tissues. However, it's up to the user to decide whether to keep or remove these genes. Also, note that this filter only affects STdeconvolve. If mitochondrial and/or ribosomal genes were not removed in the **QC & data transformation** module, they will remain in the dataset for other analyses.

Next, specify the number of top variable genes by entering "3000" in the **Use this many variable genes** text box. Selecting more genes improves the accuracy of the fit but also increases execution time. For Visium data, users might also try using more genes (e.g., 5000), in order to attempt more specific cell types. With all required parameters set, click the **RUN LDA MODELS** button to start the analysis.



Computationally intensive analyses, such as LDA model fitting, can take time. In spatialGE, users have the option to receive an email notification once the analysis is complete. To receive a notification, click the "Yes" radio button (green arrow) below the **Send email when completed?** prompt.



After the model fitting is completed, a series of plots (one per sample) are displayed under the **RUN LDA MODELS** button. These plots summarize the fit of the LDA models to the data and are crucial for selecting the model to be used during the biological identification of the topics.

The plots show three metrics:

- Perplexity (red line): Measures how well topics represent the data. Lower perplexity indicates a better fit.
- Number of rare topics (blue line): Indicates the number of topics that appear in only a few spots. While an increase in rare topics can suggest overfitting, it is not always detrimental. However, many rare topics

usually contribute little to understanding the overall tissue architecture. Therefore, it is recommended to keep the number of rare topics to a minimum.

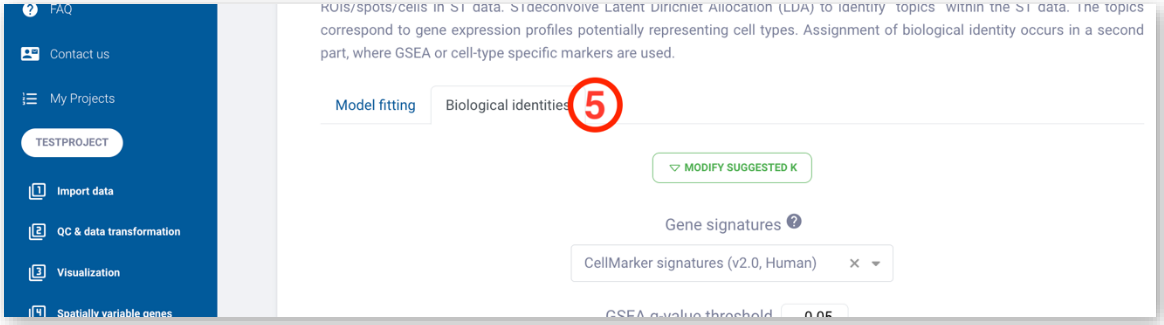
- Model alpha (gray/orange numbers in parentheses on the x-axis): Measures how different each topic is from the others. Ideally, alpha should be close to 1. In models with alpha >> 1, topics are informative as they tend to contain similar information, failing to capture differences among cell types.

Based on these metrics, spatialGE makes attempts to recommend the best model, indicated above the plot for each sample. For example, "*Suggested K=10*" (turquoise arrow) means that the LDA model with 10 topics is recommended. The user is encouraged to examine carefully this recommendation. A different model can be selected in the next part of the analysis. Saving the results from different models and comparing them can help decide which model best reflects the biology in the samples.

STdeconvolve – Biological Identities

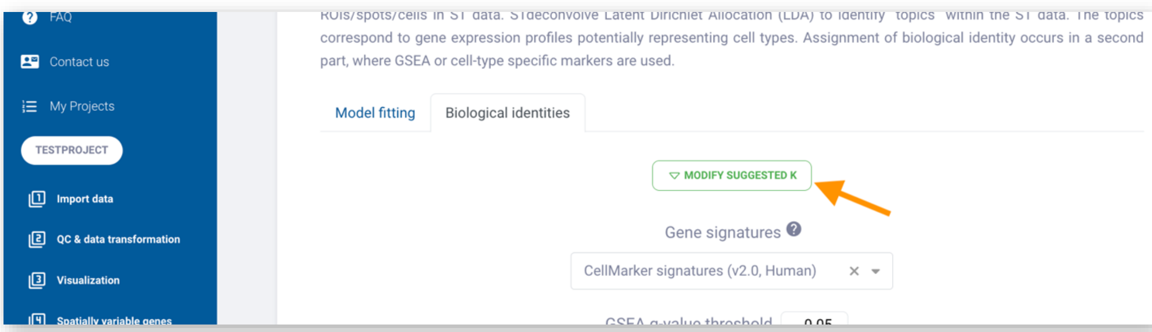
Once LDA models are fitted, users can run GSEA on the gene expression of the topics in the best-fitting model to assign a biological identity to each topic. The spatialGE app includes built-in gene sets for cell types in several cancer and tissue types. In the future, users will be able to upload custom gene sets that reflect more appropriately the biology of their study systems.

To start identifying the biological topics, click on the **Biological identities** tab that appears above the parameter controls.

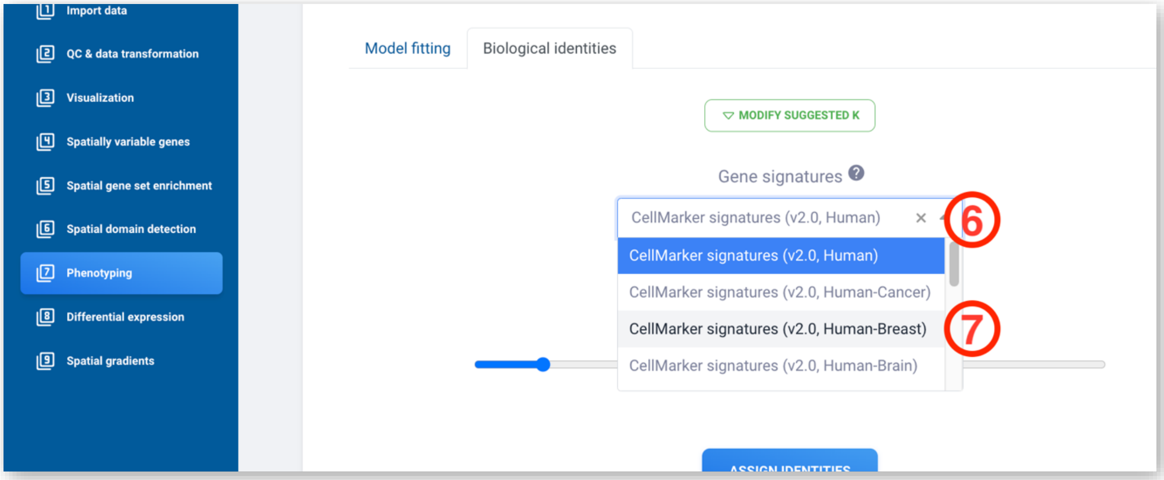


The first element in the **Biological identities** tab is the **MODIFY SUGGESTED K** button (orange arrow). Although no modification will be made to the automatic suggestions, if the user considers that the K suggested in the previous step does not correctly take into account the model fitting parameters (perplexity, number of rare topics, and alpha), or it does not represent the tissue biology (e.g., number of cell types), a different number can be entered for one or several samples in the data set.

The first element in the **Biological identities** tab is the **MODIFY SUGGESTED K** button (orange arrow). While no changes will be made to the automatic suggestions in this tutorial, users can enter a different number for one or more samples in the dataset if they believe the suggested K from the previous step does not accurately reflect the model fitting parameters (such as perplexity, number of rare topics, and alpha) or the tissue biology (e.g., number of cell types).

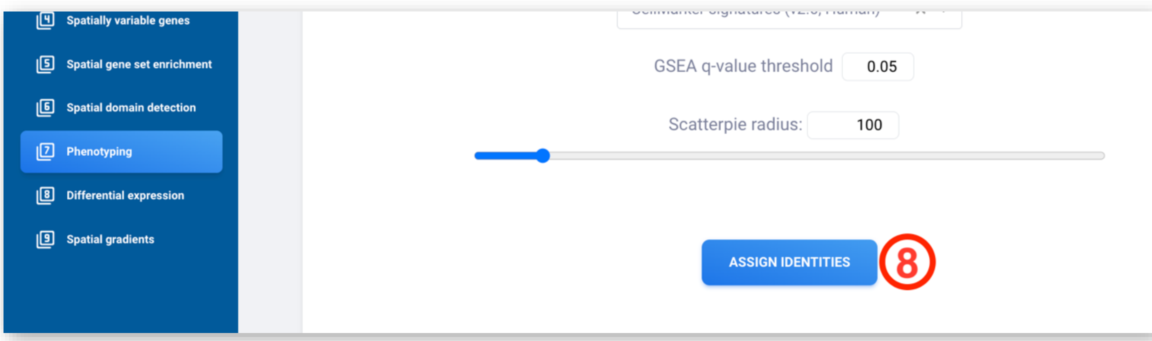


For now, click on the **Gene signatures** dropdown menu and select the *CellMarker signatures (v2.0, Human-Breast)* option. These gene sets are sourced from the 2 CellMarker 2.0 database ([Hu et al. 2023]()).

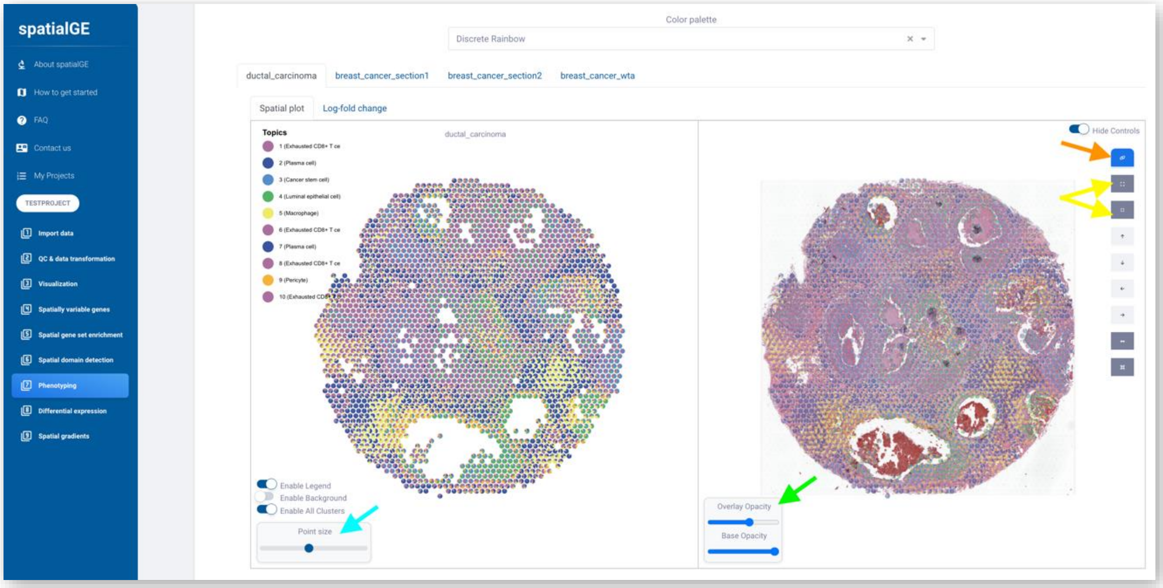


The **GSEA q-value threshold** parameter will remain at its default value. This parameter controls the q-value at which a gene set score is considered enriched. GSEA scores with q-values higher than this threshold will not be considered.

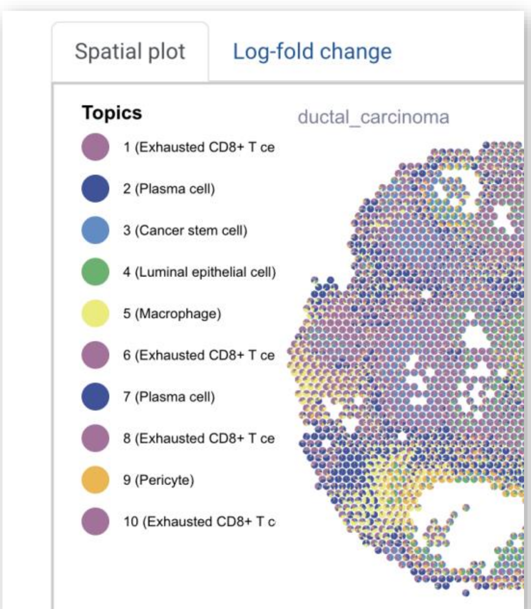
To start the analysis, click the **ASSIGN IDENTITIES** button.

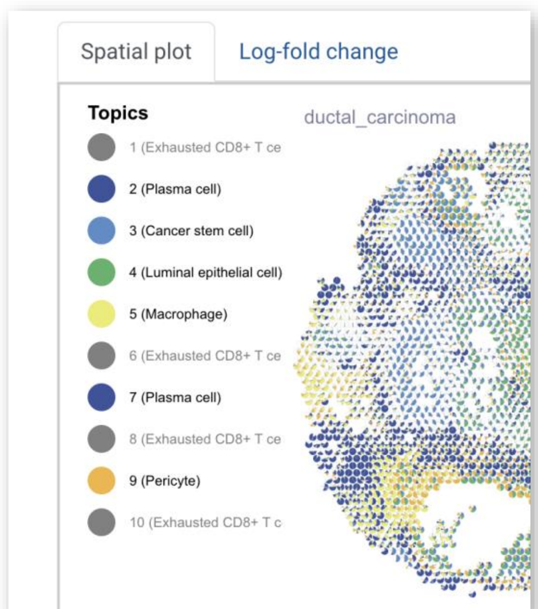


After completion, the results are displayed in individual tabs, one for each sample. Within each tab, there are two additional sub-tabs. The first sub-tab, named **Spatial plot**, presents the relative estimated abundance of each LDA topic at each spot in the sample, also known as a "scatterpie". The resulting scatterpies can be modified using several controls in the interface. Some of the most important controls include "Point size" to modify the size of the pie charts (turquoise arrow), opacity of the plot and tissue image (if available, green arrow), and adjustment of plot to tissue image (manual image registration, yellow arrow). Finally, images can be exported as PDF/PNG/SVG first by clicking the "Synchronize" button (orange arrow) and then the "Export" button (three-bar icon appearing after clicking the "Synchronize" button).



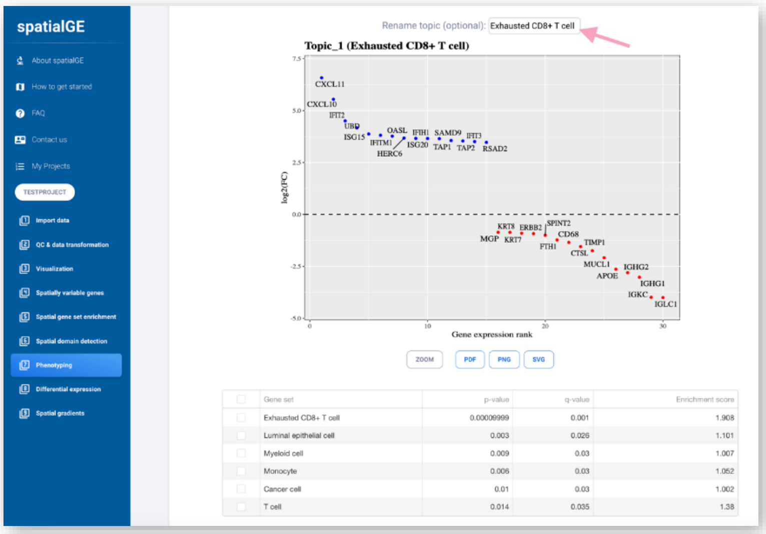
Clicking a topic color in the legend hides the corresponding portions of each pie chart in the plot (for example, “1 (Exhausted CD8+…”):





The second tab (**Log-fold change**) contains a series of plots, one per topic, showing the genes with the highest and lowest log-fold change. This is the log-fold change of a gene in a given topic compared to others. Following each plot, a table is presented with the results of the GSEA.

The second sub-tab within each sample's tab, called **Log-fold change**, features a series of plots, each representing a different topic. These plots highlight the genes with the highest and lowest log-fold changes. The log-fold change indicates how much a gene's expression varies in a specific topic compared to others. Below each plot, a table displays the results of the enrichment analysis. In this sub-tab, users can also modify the name assigned to each topic by GSEA if they believe that genes with the highest (or lowest) log-fold change represent a different cell type. This manual curation step is a common practice during cell phenotyping tasks. To make changes, enter a different name in the **Rename topic** textbox (indicated by the pink arrow)



After a change in the topic name has been made, a new button labeled **RENAME TOPICS** will appear (purple arrow). We recommend making all changes across all topics and samples before clicking the **RENAME TOPICS** button.

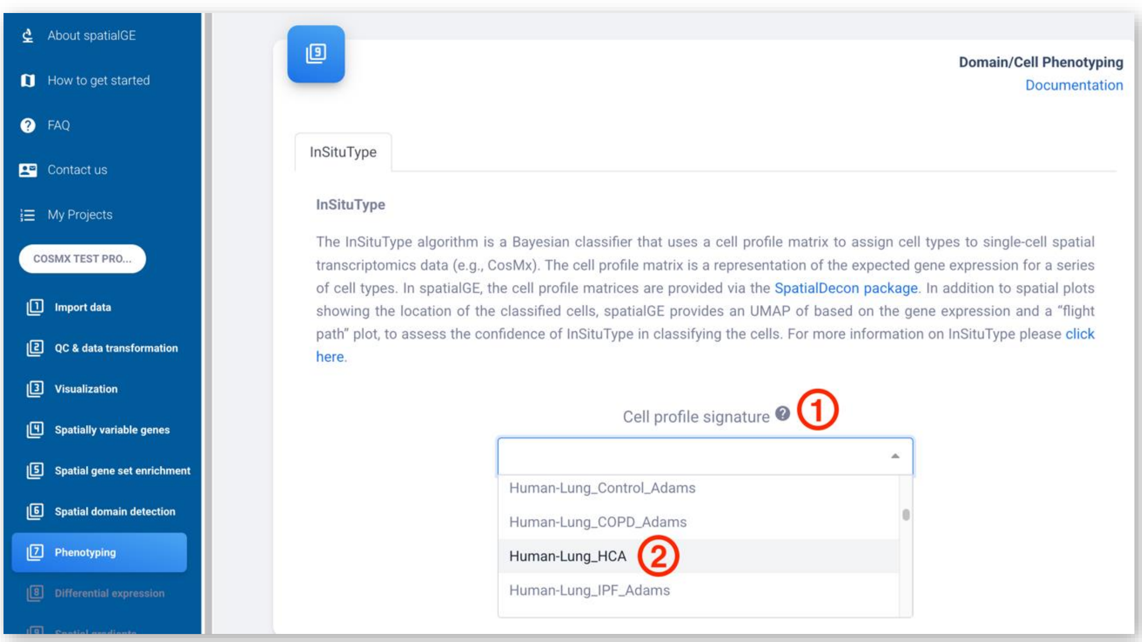
InSituType

The spatialGE web application integrates InSituType (Danaher et al. 2022) to identify cell types in single-cell-level spatial transcriptomics (ST). The InSituType algorithm uses a likelihood-based approach to weigh the gene expression information and avoids the use of dimensionality reduction techniques. The algorithm matches the ST data to a reference cell profile and provides a confidence estimate that each cell in the ST data belongs to a group/cell type in the reference data set.

Note: The InSituType algorithm is only enabled in spatialGE if *CosMx-SMI* was the selected platform during project creation.

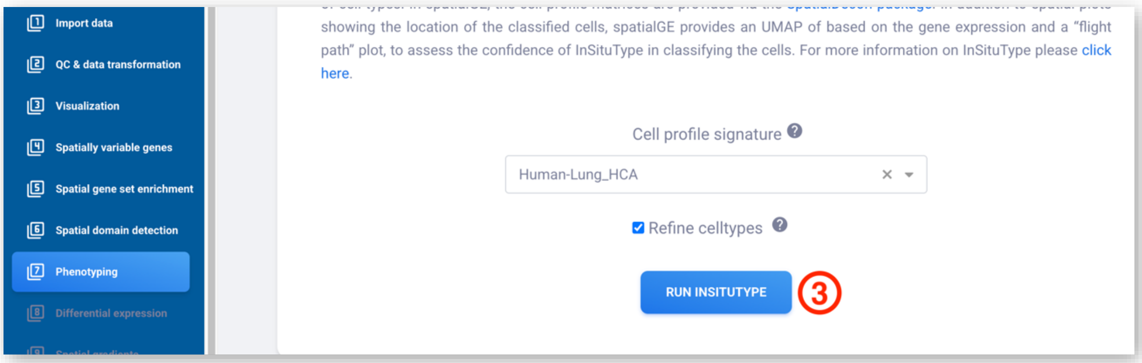
The InSituType tab is enabled in the **Phenotyping** module after normalization in QC & data transformation has been performed. For this tutorial, the CosMx demo data set will be used, which can be accessed via the "How to get started" section in the sidebar. The demo data set contains a series of fields of vision (FOVs) from two non-small-cell lung cancer (NSCLC) tissues profiled with CosMx-SMI. More information about the data set can be found here.

In the **Phenotyping** module, select "Human-Lung_HCA" from the **Cell profile signature** dropdown. In spatialGE, the cell profile signatures are provided via the SpatialDecon R package.

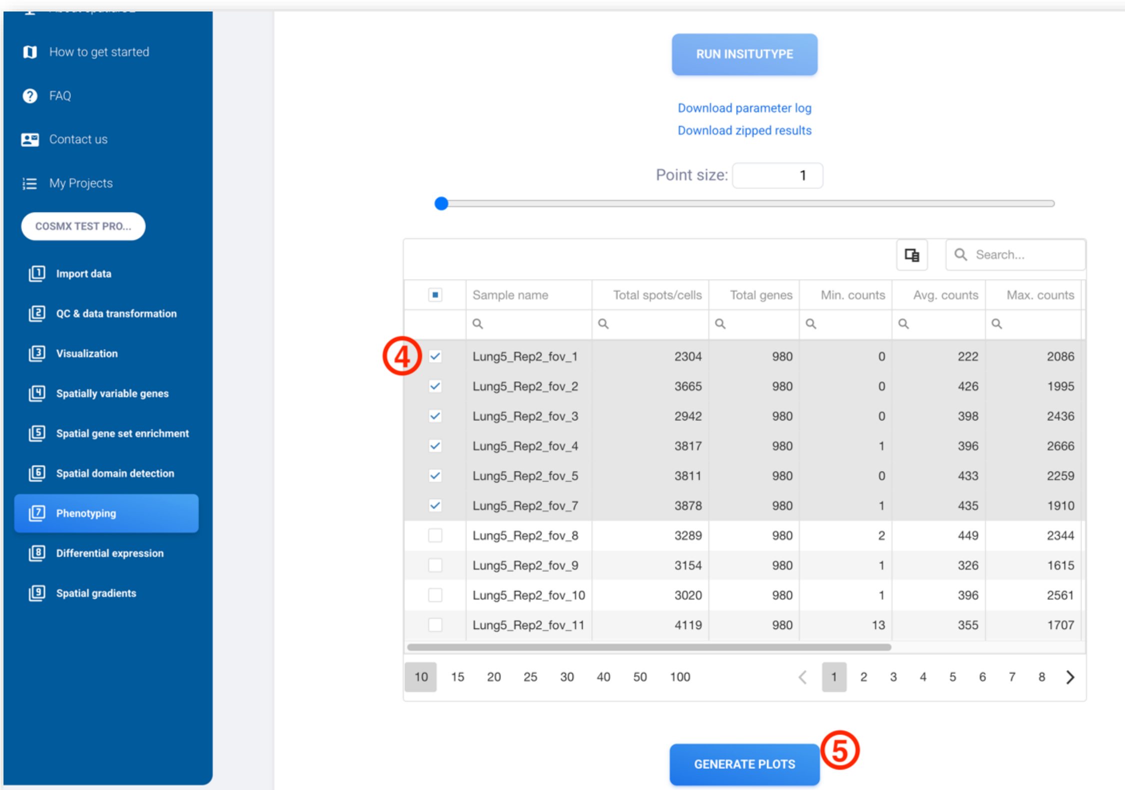


The **Refine cell types** checkbox adds an extra step to the InSituType algorithm, where log-likelihoods are re-computed within each cell type assigned during a first pass. It is generally recommended that this checkbox be kept active.

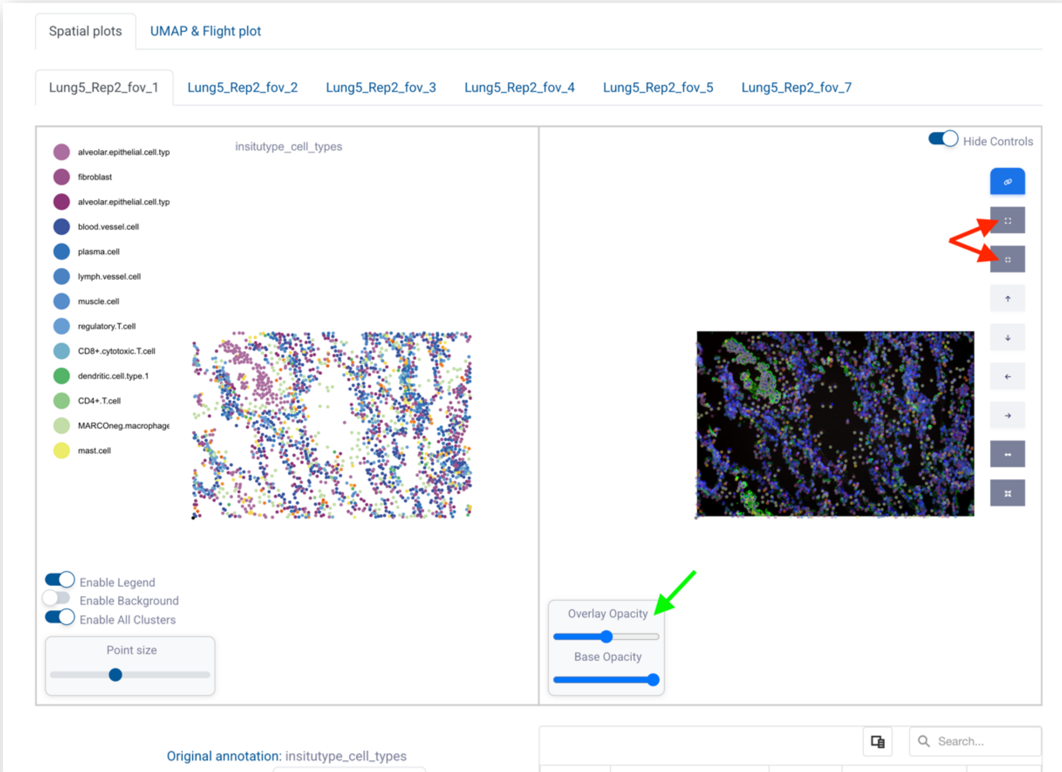
To start InSituType, click **RUN INSITUTYPE**. The process might take some time, but users can request an email notification when it is completed.



After InSituType is completed, a table with FOV IDs is presented. Users can select FOVs using the checkboxes in the first column of the table. The identified cell types will be plotted for the selected FOVs. After FOV selection, click **GENERATE PLOTS**.



The interface contains a number of controls to adjust the opacity of the plot and tissue image (if available, green arrow), and adjustment of plot to tissue image (manual image registration, red arrow).



Additionally, results from differential expression analysis among cell types are shown in table format. Importantly, the differential expression analysis within the InSituType implementation is conducted only for the top 50 most variable genes. Users must use the **Differential expression** module for a complete analysis of the identified cell types. The InSituType implementation also allows manual renaming of the identified cell types (turquoise arrows). The name of the new manual annotation can be also changed in the "Modified annotation" text box (pink arrow). Make sure to save the changes of the manual annotation ("SAVE CHANGES" located at the bottom of the page).